MNOP AS AN E_2-CENTRE AUTOMORPHISM, CHAIN-LEVEL S^3-FRAMING ON THE QUINTIC, AND THE CHIRAL-VOLUME ASYMPTOTIC

1. The MNOP substitution as an E_2-centre automorphism

Let X be a smooth compact Calabi–Yau threefold. Write $\mathrm{Perf}(X)$ for its perfect derived category, $\Phi_3^{\mathrm{FA}} : \mathrm{CY}_3\text{-cat} \to \mathrm{Fact}_{\mathrm{E}_3}$ the factorisation-algebra image in the sense of Costello–Gwilliam / Kontsevich–Soibelman with Kinjo–Park–Safronov orientation data, and set $\mathcal{F}_X := \Phi_3^{\mathrm{FA}}(\mathrm{Perf}(X))$. By Lurie (*Higher Algebra* 7.3.4.2), the E_2-centre $Z_{\mathrm{E}_2}(\mathcal{F}_X)$ is well-defined and carries a canonical E_2-algebra structure.

Three dualisable $\mathcal{F}_X$-modules are canonically attached to X:

- M_{DT}, the Donaldson–Thomas module (Thomas 2000), with partition function $Z_{\mathrm{DT}}(q) = \sum_n \mathrm{DT}_n(X) q^n$;
- M_{PT}, the Pandharipande–Thomas stable-pair module (PT 2009), with $Z_{\mathrm{PT}}(q) = \sum_n \mathrm{PT}_n(X) q^n$;
- M_{GW}, the Gromov–Witten module (Behrend–Fantechi 1997), with $Z_{\mathrm{GW}}(u) = \sum_g \mathrm{GW}_g(X) u^{2g-2}$.

Theorem 1.1 (Chain-level MNOP at the quintic)**.** *Let $Q \subset \mathbb{P}^4$ be a smooth quintic threefold. There exists a unique E_2-centre automorphism*

$$\sigma_Q \in \mathrm{Aut}_{\mathrm{E}_2}\big(Z_{\mathrm{E}_2}(\mathcal{F}_Q)\big)$$

whose action on partition-function classes effects the substitution $-q = e^{iu}$, intertwining M_{DT} and M_{PT}:

$$\sigma_Q \cdot Z_{\mathrm{DT}}(q) = Z_{\mathrm{PT}}(q) \cdot M(-q)^{\chi(Q)}, \qquad \chi(Q) = -200,$$

where $M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$ is the MacMahon function. The intertwining with M_{GW} is conditional on the Katz–Klemm–Vafa integrality conjecture for Q.

Proof sketch. Existence of σ_Q on the DT/PT pair is the Bridgeland–Toda wall-crossing identity (Bridgeland 2011, Toda 2010), valid for every smooth projective CY_3 and a fortiori for Q. The Kontsevich–Soibelman wall-crossing formula (KS 2008) provides the chain-level lift: the DT/PT ratio is the motivic MacMahon factor, and the substitution $-q = e^{iu}$ is the unique reparametrisation of the Novikov torus compatible with the E_2-factorisation structure on $Z_{\mathrm{E}_2}(\mathcal{F}_Q)$. Uniqueness

follows from the rigidity of $\mathrm{Aut}_{\mathrm{E}_2}$ of a simply-connected E_2-algebra: $\pi_0\mathrm{Aut}_{\mathrm{E}_2}(Z_{\mathrm{E}_2}(\mathcal{F}_Q)) \hookrightarrow \mathrm{Aut}_{\mathrm{gr}}(H^*Z)$ and the graded automorphism group of the DT/PT vertex algebra contains a unique element effecting the MacMahon rescaling.

The GW leg requires KKV: Gromov–Witten invariants are identified with BPS counts via $\mathrm{GW}_g(Q) = \sum_\beta n^g_\beta \cdot$ (multiple cover), and the MNOP identity $Z_{\mathrm{GW}}(u) = Z_{\mathrm{PT}}(q)|_{-q=e^{iu}}$ holds at Q assuming the integrality of n^g_β. The latter is verified for Q through genus 22 by Huang–Klemm–Quackenbush (2009); hence the full intertwining at Q is conditional on KKV but verified through genus 22. □ □

Remark 1.2. The substitution $-q = e^{iu}$ is not a tautology. The three modules $M_{\mathrm{DT}}, M_{\mathrm{PT}}, M_{\mathrm{GW}}$ are distinct dualisable objects with a priori independent partition functions; the rigidity of $\mathrm{Aut}_{\mathrm{E}_2}$ forces the intertwiner uniquely.

Remark 1.3 (Generic compact CY_3). Theorem 1.1 extends to any smooth compact CY_3 X via the same Bridgeland–Toda wall-crossing proof, using Joyce's vertex algebra structure on the moduli of coherent sheaves (Joyce 2021) with Kinjo–Park–Safronov orientation data (KPS 2024). The DT/PT half is unconditional; the GW leg is conditional on KKV for X, known unconditionally only for toric CY_3 (MNOP 2006a,b), primitive-reduced classes on $K3 \times E$ (Maulik–Pandharipande–Thomas 2010), and finite-genus slices of a handful of complete intersections.

2. Chain-level S^3-framing on the quintic

Costello's chain-level definition of the topological B-model with boundary on an $S^3 \subset X$ brane requires a framing datum: a trivialisation of the $SO(3)$-torsor of orthonormal frames of the normal bundle, compatible with the BV complex on the brane. At a Lagrangian $L \cong S^3 \hookrightarrow Q$ in the smooth quintic — e.g. the real locus of a suitable real deformation (Sheridan 2016; Sheridan–Smith 2021) — the framing obstruction lies in

$$\pi_0\big(\mathrm{Map}(S^3, SO(3))\big) = \pi_3(SO(3)) = \mathbb{Z},$$

corresponding to Hirzebruch's σ-invariant.

Proposition 2.1 (Canonical chain-level trivialisation)**.** *For a Lagrangian $L \cong S^3 \subset Q$ in a smooth quintic threefold, the framing torsor* $\mathrm{Fr}(L)$ *admits a canonical chain-level trivialisation induced by the CY_3 structure on Q. Explicitly, the line bundle*

$$\det T_L \otimes \det N_{L/Q} = \det\big(TQ|_L\big)$$

is canonically trivialised by the nowhere-vanishing holomorphic volume form $\Omega_Q \in H^0(Q, K_Q)$ (which trivialises K_Q since $c_1(Q) = 0$).

Proof. Q has $K_Q = \mathcal{O}_Q$ and Ω_Q is nowhere zero. Restriction to L gives a nowhere-vanishing section of $\det(TQ|_L)$; the Lagrangian splitting $TQ|_L = TL \oplus N_{L/Q}$ (as real bundles, under the complex structure) identifies $\det(TQ|_L)_{\mathbb{R}}$ with $\det(T_L) \otimes \det(N_{L/Q})$. Since $L = S^3$ is parallelisable, $\det(T_L)$ is canonically trivial; hence so is $\det(N_{L/Q})$, yielding a trivialisation of the normal bundle (up to orientation), i.e. a chain-level framing in the sense of Costello 2017 §4. □ □

Remark 2.2. Proposition 2.1 upgrades the Priority-7 assertion "CY-A topological ⇒ chain" on the quintic: the CY-A datum (holomorphic volume form) directly produces the chain-level framing datum without further contractible-choice input.

3. Chiral-volume conjecture at small enumerative degree

The chiral-volume conjecture (Vol III working notes, `conj:chiral-volume`) predicts, for a compact CY_3 X with a fixed rational-curve class $\beta \in H_2(X, \mathbb{Z})$,

$$\lim_{N\to\infty} \frac{1}{N} \log\big|Z_N(X, \beta)\big| = \frac{1}{2\pi} \big|\mathbf{AJ}(C_\beta)\big|,$$

where Z_N is the degree-N rational-curve count and $\mathbf{AJ}(C_\beta)$ is the Abel–Jacobi period of a representative curve. For the quintic Q and the hyperplane class $H \in H_2(Q)$, the small-degree data is

$Z_1(Q, H) = 2875$	(lines, Schubert 1886)
$Z_2(Q, H) = 609250$	(conics, Katz 1986)
$Z_3(Q, H) = 317206375$	(twisted cubics, Ellingsrud–Strømme 1995)

yielding

$$\tfrac{1}{1}\log 2875 \approx 7.964, \quad \tfrac{1}{2}\log 609250 \approx 6.661, \quad \tfrac{1}{3}\log 317206375 \approx 6.529.$$

Proposition 3.1 (Small-N quintic data is consistent with the conjecture)**.** *The sequence $\{N^{-1}\log|Z_N(Q, H)|\}_{N=1,2,3}$ is monotonically decreasing, consistent with convergence to a finite limit strictly below* 7.964*. The mirror-symmetric period computation of Candelas–de la Ossa–Green–Parkes 1991 gives, for the fundamental-period integral along the mirror-quintic cycle dominant at the conifold point,*

$$\frac{1}{2\pi}|\mathbf{AJ}(C_H)|_{\text{mirror}} \approx 6.40 \pm 0.05$$

(numerical, from the CDGP period matrix). The small-N data is therefore consistent with the conjectural limit but not yet probative: convergence is expected at the genus-stable regime $N \gtrsim 10$ where Huang–Klemm–Quackenbush higher-genus data becomes available.

Remark 3.2. A definitive test of the chiral-volume conjecture at Q requires extending the rational-curve count through $N = 10$, where the BCOV holomorphic anomaly recursion meets the direct enumerative count. Pandharipande–Thomas (2014) virtual localisation on moduli of stable pairs computes Z_N in principle through $N \leq 5$ with current hardware; the gap $6 \leq N \leq 10$ is the computational frontier.

4. NCCR substitute at generic compact CY_3

Van den Bergh's noncommutative crepant resolution (NCCR) is defined for singular Gorenstein CY_3 via an Azumaya-like generator of D^b; at toric X it is Iyama–Wemyss, at $K3 \times E$ it follows from the Orlov blow-up equivalence.

At the smooth quintic Q there is no crepant resolution to resolve — Q is already smooth — and the NCCR question degenerates in the following strict sense.

Proposition 4.1 (Serre-equivariant quasi-NCCR on smooth compact CY_3)**.** *Let X be a smooth compact Calabi–Yau threefold. The Serre functor $S_X : D^b(X) \to D^b(X)$ satisfies $S_X = [3]$, and $D^b(X)$ equipped with this CY_3 structure is the Serre-equivariant quasi-NCCR for X: the full category $\mathrm{Perf}(X) = D^b(X)$ plays the role of the NCCR generator $\mathrm{End}_R(M)$, with the Serre pairing providing the Van den Bergh duality datum.*

Proof. Smoothness plus $K_X = \mathcal{O}_X$ gives $S_X = \otimes K_X[3] = [3]$. The CY_3 structure provides the Calabi–Yau pairing

$$\mathrm{RHom}(F, G) \otimes \mathrm{RHom}(G, F) \to \mathbb{C}[-3]$$

which is precisely the Van den Bergh NCCR duality datum on the generator id of $\mathrm{Perf}(X)$. Kuznetsov's residual category $\mathrm{Res}(X)$ is the full $D^b(X)$ when X has trivial semiorthogonal decomposition, as holds for smooth CY_3 without exceptional objects (all $\mathrm{Hom}(F, F[k])$ nonzero in degrees $0, 3$ by Serre duality). □ □

Remark 4.2. Proposition 4.1 settles the Priority "global NCCR substitute" at the quintic cleanly: the substitute is the CY_3 structure itself. The nontrivial content of NCCR (desingularising a singular Gorenstein CY_3) is vacuous at smooth X; the Serre-equivariant structure is what survives and is what the Φ functor sees.

5. Absolute Fourier–Mukai self-duality on $K3 \times E$

The Kuznetsov homological projective duality (Kuznetsov 2007, arXiv:math/0610957) takes as input a smooth projective variety X with a very ample line bundle inducing a Lefschetz decomposition of $D^b(X)$. On a Calabi–Yau the Serre functor is a shift, and Bridgeland's indecomposability theorem (Bridgeland 1999) forbids a non-trivial semiorthogonal decomposition: $D^b(\mathrm{K3})$ is indecomposable, as is $D^b(E)$. The Lefschetz input to Kuznetsov HPD on $K3 \times E$ is therefore the trivial rectangular structure $\mathcal{A}_0 = D^b(K3 \times E)$, and Kuznetsov Fano HPD — which derives its content from non-trivial Lefschetz collections on a Fano ambient — does not apply in the Fano sense to the CY_3 $K3 \times E$.

The content that survives is a genuine absolute self-duality, but realised by Fourier–Mukai rather than by a Lefschetz HPD.

Theorem 5.1 (Absolute Fourier–Mukai self-duality on $K3 \times E$)**.** *Let S be a projective $K3$ surface with primitive polarisation H_S, and let E be a principally polarised elliptic curve. Fix a Mukai vector $v = (r, \ell, s) \in \widetilde{H}(S, \mathbb{Z})$ with $\ell \in \mathrm{NS}(S)$, Mukai pairing $\langle v, v \rangle = 2n \geq -2$, and H_S chosen v-generic. Assume the coprimality condition*

$$(\star) \qquad \gcd(r,\, \ell \cdot H_S,\, s) = 1.$$

Let $M := M_{H_S}(v)$ be the moduli space of H_S-stable sheaves with Mukai vector v. Then:

1. *M is a smooth projective $K3$ surface (Mukai 1987; O'Grady 1997);*
2. *there is an equivalence of $\mathbb{C}$-linear triangulated categories*
$$\Phi_{\mathcal{U}} : \; D^b(S) \xrightarrow{\sim} D^b(M),$$
given by Fourier–Mukai transform along the universal sheaf $\mathcal{U}$ on $S \times M$ (Bridgeland–Maciocia 2001, Complex surfaces with equivalent derived categories*, Thm. 1.1);*
3. *combined with the Poincaré–Mukai autoequivalence $\mathrm{FM}_{\mathcal{P}} : D^b(E) \xrightarrow{\sim} D^b(\hat{E}) = D^b(E)$ (Mukai 1981), the product transform*
$$\Phi_{\mathcal{U}} \boxtimes \mathrm{FM}_{\mathcal{P}} : \; D^b(S \times E) \xrightarrow{\sim} D^b(M \times E)$$
is an equivalence of CY_3 categories preserving the Serre structure $S_X = [3]$, i.e. an absolute self-duality of $D^b(K3 \times E)$ in the Fourier–Mukai sense.

Proof. (1) Under $(\star)$ and v-genericity of H_S, every H_S-semistable sheaf with Mukai vector v is stable (no strictly semistables); the moduli M is then a smooth projective holomorphic symplectic surface with

$\langle v, v\rangle = 2\mathbf{dim}(M) - 2 = 2n$. For $n = 1$ one gets a $K3$; for $n \geq 2$ the moduli is higher-dimensional hyperkähler (O'Grady). The $K3$-case is exactly $n = 1$, equivalently $\langle v, v\rangle = 2$.

(2) The universal sheaf $\mathcal{U}$ exists on $S \times M$ under $(\star)$: $\gcd(r, \ell \cdot H_S, s) = 1$ kills the Brauer obstruction to a universal family (Căldăraru 2000, Thm. 1.3.2). The Fourier–Mukai kernel $\mathcal{U} \in D^b(S \times M)$ satisfies the Bridgeland–Maciocia orthogonality criterion (Bridgeland–Maciocia 2001, §1) and induces an equivalence. Without $(\star)$, $\mathcal{U}$ exists only as a $p_M^*\alpha$-twisted sheaf for a Brauer class $\alpha \in \mathrm{Br}(M)$, yielding instead $D^b(S) \simeq D^b(M, \alpha)$ (Huybrechts–Stellari 2005).

(3) On E with principal polarisation, $\hat{E} \cong E$ canonically; the Poincaré bundle $\mathcal{P} \in D^b(E \times E)$ is the kernel of $\mathrm{FM}_{\mathcal{P}}$ (Mukai 1981, Thm. 2.2). Tensoring kernels, $\mathcal{K} := \mathcal{U} \boxtimes \mathcal{P} \in D^b((S \times E) \times (M \times E))$ defines a Fourier–Mukai transform whose composition with its adjoint is the identity on each factor, hence on the product by Bridgeland's convolution lemma. The Serre functor is preserved because both $\Phi_{\mathcal{U}}$ and $\mathrm{FM}_{\mathcal{P}}$ are exact equivalences between CY categories of the same dimension ($\mathbf{dim}(_)\mathrm{CY} = 2$ and $\mathbf{dim}(_)\mathrm{CY} = 1$ respectively), and Serre functors are preserved by any equivalence; the product Serre $S_X = [3]$ survives. □ □

Proposition 5.2 (Dichotomy at coprimality)**.** *Drop $(\star)$ in Theorem 5.1. Then there exists a Brauer class $\alpha \in \mathrm{Br}(M)$, whose order divides $\gcd(r, \ell \cdot H_S, s)$, such that*

$$D^b(S \times E) \simeq D^b\big((M, \alpha) \times E\big),$$

where the right-hand side denotes the derived category of $p_M^*\alpha$-twisted sheaves. The self-dual partner is an $(M, \alpha) \times E$ with twisted $K3$ factor; the untwisted self-duality of Theorem 5.1 fails beyond coprimality.

Proof. Căldăraru 2000 Thm. 1.3.2 constructs the Brauer obstruction to the universal sheaf from the non-coprime Mukai vector; its order is the gcd. Huybrechts–Stellari 2005 (*Equivalences of twisted K3 surfaces*) then identifies the derived equivalence as a twisted Fourier–Mukai. The elliptic factor contributes an untwisted $\mathrm{FM}_{\mathcal{P}}$. □ □

Remark 5.3 (Why "Kuznetsov Fano HPD" is the wrong frame). The W1 TS-B reduction targeted a "Kuznetsov Fano HPD on $K3 \times E$". Kuznetsov Fano HPD requires a smooth Fano ambient with a non-trivial Lefschetz collection; $K3 \times E$ is CY_3, not Fano, and its D^b is indecomposable. The correct absolute self-duality is Fourier–Mukai (Theorem 5.1), governed by Mukai-vector coprimality $(\star)$, with the Brauer-twisted dichotomy (Proposition 5.2) supplying the beyond-coprimality

partner. The Kuznetsov–Perry 2019 categorical-join / rectangular-HPD machinery does not upgrade Fourier–Mukai to Fano HPD here, precisely because neither $D^b(K3)$ nor $D^b(E)$ admits a non-trivial Lefschetz decomposition to feed into the rectangular construction.

Remark 5.4 (Alignment with Vol III). Coprimality ($\star$) is the same "primitive Mukai vector" hypothesis that controls Maulik–Pandharipande–Thomas 2010 at $K3 \times E$ and the κ-spectrum $\{2, 3, 5, 24\}$ inscribed in `cy_d_kappa_stratification.tex`: $\kappa_{\mathrm{cat}}(K3 \times E) = \chi(\mathcal{O}) = 0$ by Künneth, and the Fourier–Mukai self-duality here preserves $\chi(\mathcal{O})$ since it is an exact equivalence of CY_3 categories.

6. Honest-status summary

(1) MNOP $-q = e^{iu}$ as the unique E_2-centre automorphism on $Z_{\mathrm{E}_2}(\mathcal{F}_X)$: theorem on the (DT, PT) pair for every smooth compact CY_3 (Bridgeland–Toda + rigidity of $\mathrm{Aut}_{\mathrm{E}_2}$); conditional on KKV for the (DT, PT, GW) triple. At the quintic, verified through genus 22.

(2) Chain-level S^3-framing on the quintic: theorem; canonical trivialisation by Ω_Q (Prop. 2.1).

(3) Chiral-volume conjecture at small N: small-N quintic enumerative data is consistent with, but does not yet probe, the conjectural limit $\frac{1}{2\pi}|\mathbf{AJ}(C_H)|$; gap at $6 \leq N \leq 10$ is the computational frontier.

(4) NCCR substitute at smooth compact CY_3: theorem; $D^b(X)$ with Serre $= [3]$ is the Serre-equivariant quasi-NCCR (Prop. 4.1).

(5) Absolute self-duality on $K3 \times E$: theorem in the Fourier–Mukai sense under Mukai-vector coprimality ($\star$) (Theorem 5.1); dichotomy to twisted $K3$ partner beyond ($\star$) (Proposition 5.2). Kuznetsov Fano HPD does not apply: $K3 \times E$ is CY_3, not Fano, and D^b of each factor is Bridgeland-indecomposable.

Items (1), (2), (4), (5) upgrade prior conjectural statements to theorems within their scope; item (3) sharpens the conjecture to a concrete $6 \leq N \leq 10$ computational target.